Bu soruda ilk kısım 4 puan son kısım 6 puan
her iki kısım da yok ise sıfır

1) $A = S\Lambda S^{-1} \Rightarrow \det A = \det(S\Lambda S^{-1})$

$\det A = \det(S)\cdot \underbrace{\det(\Lambda)}_{\lambda_1\cdot\lambda_2\cdots\lambda_n} \det(S^{-1})$

4

$\Rightarrow \det A = \lambda_1\cdot\lambda_2\cdot\lambda_3\cdots\lambda_n \cdot \underbrace{\det(SS^{-1})}_{1}$

$\det A = \lambda_1\lambda_2\cdots\lambda_n \Rightarrow$

Sadece bu denklem yazılmış ise puan yok...

6 if $\det A > 1 \Rightarrow$ at least one eig. value must be ≥ 1 and if an eig. value > 1

$\Rightarrow$ System is astable.

2) $f(x)=1,\ g(x)=x^2,\ h(x)=x^4$

chose a 1st order function $O(x) = ax^5+bx^4+cx^3+dx^2+ex+f$

add if $O(x) \perp f(x) \Rightarrow \int_0^1 O(x)f(x)dx = 0 \quad \frac{a}{6}+\frac{b}{5}+\frac{c}{4}+\frac{d}{3}+\frac{e}{2}+f=0$

10 if $O(x) \perp g(x) \Rightarrow \int_0^1 O(x)g(x)dx = 0 \quad \frac{a}{8}+\frac{b}{7}+\frac{c}{6}+\frac{d}{5}+\frac{e}{4}+\frac{f}{3}=0$

if $O(x) \perp h(x) \Rightarrow \int_0^1 O(x)h(x)dx = 0 \quad \frac{a}{10}+\frac{b}{9}+\frac{c}{8}+\frac{d}{7}+\frac{e}{6}+\frac{f}{5}=0$

5

$$\underbrace{\begin{bmatrix} 1/6 & 1/5 & 1/4 & 1/3 & 1/2 & 1 \\ 1/8 & 1/7 & 1/6 & 1/5 & 1/4 & 1/3 \\ 1/10 & 1/9 & 1/8 & 1/7 & 1/6 & 1/5 \end{bmatrix}}_{A} \cdot \begin{bmatrix} a \\ b \\ c \\ d \\ e \\ f \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 6/5 & 6/4 & 2 & 3 & 6 \\ 1 & 8/7 & 8/6 & 8/5 & 2 & 8/3 \\ 1 & 10/9 & 10/8 & 10/7 & 10/6 & 2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 6/5 & 6/4 & 2 & 3 & 6 \\ 0 & \frac{-2}{35} & -1/6 & -2/5 & -1 & \frac{-10}{3} \\ 0 & \frac{-4}{45} & -1/4 & -4/7 & \frac{-8}{6} & -4 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 6/5 & 6/4 & 2 & 3 & 6 \\ 0 & -2 & -35/6 & -70/5 & -35 & -350/3 \\ 0 & 0 & \frac{10}{24} & \frac{16}{7} & 10 & \frac{160}{3} \end{bmatrix} \leftarrow \begin{bmatrix} 1 & 6/5 & 6/4 & 2 & 3 & 6 \\ 0 & -2 & -35/6 & -70/5 & -35 & \frac{-350}{3} \\ 0 & -4 & -45/4 & \frac{-180}{7} & \frac{-120}{2} & -180 \end{bmatrix}$$

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$d=1, e=0, f=0$ (Free)

$\frac{10}{24}c + \frac{16}{7}d = 0 \Rightarrow c = \frac{-24.16}{10.7} = -5.5$

$2b + \frac{35}{6}c + \frac{70}{5}d = 0 \Rightarrow b = 9$

$a + \frac{6}{5}b + \frac{6}{4}c + 2d = 0 \Rightarrow a = -4.6$

$$O(x) = -4.6x^5 + 9x^4 - 5.5x^3 + x^2$$

5 The nullspace of A is 3-Dimensional (3 Free cols) $\Rightarrow$ Infinite functions can be found.

8 3. quadratic function: $f(t) = at^2 + bt + c$

March $\Rightarrow t=1 \Rightarrow a+b+c=20$
April $\Rightarrow t=2 \Rightarrow 4a+2b+c=40$
May $\Rightarrow t=3 \Rightarrow 9a+3b+c=10$

$$\Rightarrow \underbrace{\begin{bmatrix}1&1&1\\4&2&1\\9&3&1\end{bmatrix}}_{A} \cdot \underbrace{\begin{bmatrix}a\\b\\c\end{bmatrix}}_{x} = \underbrace{\begin{bmatrix}20\\40\\10\end{bmatrix}}_{b}$$

8 $A^TAx = A^Tb$

$$\begin{bmatrix}1&4&9\\1&2&3\\1&1&1\end{bmatrix}\cdot\begin{bmatrix}1&1&1\\4&2&1\\9&3&1\end{bmatrix}\cdot\begin{bmatrix}a\\b\\c\end{bmatrix} = \begin{bmatrix}1&4&9\\1&2&3\\1&1&1\end{bmatrix}\cdot\begin{bmatrix}20\\40\\10\end{bmatrix} \Rightarrow \begin{bmatrix}98&36&14\\36&14&6\\14&6&3\end{bmatrix}\cdot\begin{bmatrix}a\\b\\c\end{bmatrix} = \begin{bmatrix}270\\130\\70\end{bmatrix}$$

4

$$\left[\begin{array}{ccc|c}98&36&14&270\\36&14&6&130\\14&6&3&70\end{array}\right] \xrightarrow[\frac{r3}{14}]{\frac{r1}{98},\frac{r2}{36}} \left[\begin{array}{ccc|c}1&0{,}37&0{,}14&2{,}76\\1&0{,}39&0{,}17&3{,}61\\1&0{,}43&0{,}21&5\end{array}\right] \xrightarrow[-r1+r3]{-r1+r2} \left[\begin{array}{ccc|c}1&0{,}37&0{,}14&2{,}76\\0&0{,}02&0{,}03&0{,}85\\0&0{,}06&0{,}07&2{,}24\end{array}\right] \xrightarrow{-3r2+r3} \left[\begin{array}{ccc|c}1&0{,}37&0{,}14&2{,}76\\0&0{,}02&0{,}03&0{,}85\\0&0&-0{,}02&-0{,}31\end{array}\right]$$

$-0.02c = -0.31 \Rightarrow \boxed{c = 15.5}$

$0{,}02b + 0{,}03c = 0{,}85 \Rightarrow \boxed{b = 19{,}25}$

$a + 0{,}37b + 0{,}14c = 2{,}76 \Rightarrow \boxed{a = -6{,}93}$

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$\Rightarrow f(t) = -6.53t^2 + 19.25t + 15.5$

$t = 4 \Rightarrow f(4) = -12$

according to information given the parity will be,

$34000 - 12000 = 22000$ on June.

4. a_n: water level in tank A in nth time interval

b_n: " " " tank B " " " "

In 10 minutes $\rightarrow 0.5 \cdot a_n$ amount of water is transferred by Pump A

$0.1 \cdot b_n$ " " " " by Pump B

%100 a_n → 20 mins.

%50 a_n → 10 mins

%100 b_n → 100 mins.

%10 b_n → 10 mins.

10 puan

Burada yanlış düşünmüş ise yada continious için doğru yazdıysa 5 puan

burada cont. için saçma olmayan hata yaptıysa 3 puan

burada saçma hata 0 puan

a) $a_{n+1} = a_n - 0.5a_n + 0.1b_n \Rightarrow$ $a_{n+1} = 0.5a_n + 0.1b_n$

$b_{n+1} = b_n - 0.1b_n + 0.5a_n \Rightarrow$ $b_{n+1} = 0.5a_n + 0.9b_n$

b) $\begin{bmatrix} a_{n+1} \\ b_{n+1} \end{bmatrix} = \begin{bmatrix} 0.5 & 0.1 \\ 0.5 & 0.9 \end{bmatrix} \cdot \begin{bmatrix} a_n \\ b_n \end{bmatrix}$

State transition matrix (A)

doğru cevap: 10 puan

Burada yanlış matrix buldu ise 5 puan

doğru cevap: 5 puan

c) $|\lambda I - A| = 0 \Rightarrow \begin{vmatrix} \lambda - 0.5 & -0.1 \\ -0.5 & \lambda - 0.9 \end{vmatrix} = 0 \Rightarrow$

$(\lambda - 0.5)(\lambda - 0.9) - 0.05 = 0$

$\lambda^2 - 1.4\lambda + 0.4 = 0$

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d) $\lambda^2 - 1.4\lambda + 0.4 = 0$

$\Delta = 1.96 - 1.6 = 0.36 \Rightarrow \lambda_{1,2} = \frac{1.4 \pm \sqrt{0.36}}{2}$

$\lambda_1 = 1, \quad \lambda_2 = 0.4$

$\lambda_1 = 1 \Rightarrow \lambda I - A = \begin{bmatrix} 0.5 & -0.1 \\ -0.5 & 0.1 \end{bmatrix} \Rightarrow \begin{bmatrix} 0.5 & -0.1 \\ -0.5 & 0.1 \end{bmatrix} \cdot X_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

doğru cevap: 10 puan
Teknik doğru fakat kullanılan matrix yanlış ise 5 puan

$X_1 = \begin{bmatrix} 1 \\ 5 \end{bmatrix}$

$\lambda_2 = 0.4 \Rightarrow \begin{bmatrix} -0.1 & -0.1 \\ -0.5 & -0.5 \end{bmatrix} \cdot X_2 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow X_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

e) $\begin{bmatrix} a_n \\ b_n \end{bmatrix} = c_1 \lambda_1 X_1 + c_2 \lambda_2 X_2 = c_1 \cdot \begin{bmatrix} 1 \\ 5 \end{bmatrix} \cdot 1^n + c_2 \begin{bmatrix} 1 \\ -1 \end{bmatrix} (0.4)^n$

doğru cevap: 10 puan (formül yazılmış olmalı ve c1 ve c2 hesaplanmış olmalı) işlem hatası çok saçma olmadıkça sorun değil
yanlış ise: 0

for $n=0 \Rightarrow \begin{bmatrix} a_0 \\ b_0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

$\begin{bmatrix} 1 \\ 0 \end{bmatrix} = c_1 \cdot \begin{bmatrix} 1 \\ 5 \end{bmatrix} + c_2 \cdot \begin{bmatrix} 1 \\ -1 \end{bmatrix} \Rightarrow \begin{matrix} 5c_1 = c_2 \\ c_1 + c_2 = 1 \end{matrix}$

$\Rightarrow c_1 = \frac{1}{6} = 0.16$

$c_2 = 0.83$

$\Rightarrow \begin{bmatrix} a_n \\ b_n \end{bmatrix} = 0.16 \cdot \begin{bmatrix} 1 \\ 5 \end{bmatrix} + 0.83 \begin{bmatrix} 1 \\ -1 \end{bmatrix} \cdot (0.4)^n$

each time interval → 20 mins. (10 mins. ON, 10 mins. OFF)

4 hours → 20 mins. × 12 → $n = 12$

$\begin{bmatrix} a_{12} \\ b_{12} \end{bmatrix} = \begin{bmatrix} 0.16 \\ 0.83 \end{bmatrix} + \begin{bmatrix} 0.83 \\ -0.83 \end{bmatrix} \cdot (0.4)^{12} = \begin{bmatrix} 0.16 + 14 \cdot 10^{-6} \\ 0.83 - 14 \cdot 10^{-6} \end{bmatrix}$

f) $n \to \infty \Rightarrow (0.4)^n \to 0$

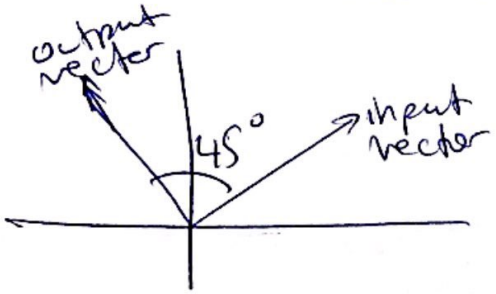
$\Rightarrow \begin{bmatrix} a_\infty \\ b_\infty \end{bmatrix} = \begin{bmatrix} 0.16 \\ 0.83 \end{bmatrix}$

(0,83 of tank B is full
0,16 of tank A is full)

doğru cevap: 5 puan

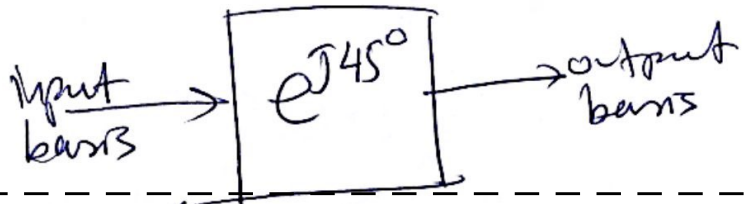
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5. 45° rotation



we should use eig. vector basis

Bu cümle 10 puan



what are the eig. vectors ↑ ?

$$\begin{bmatrix} \cos 45° & -\sin 45° \\ \sin 45° & \cos 45° \end{bmatrix} = \begin{bmatrix} \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{bmatrix} = A$$

Defterinizde zaten olan bu kısım için 5 puan

$$\begin{vmatrix} \frac{\sqrt{2}}{2}-\lambda & -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2}-\lambda \end{vmatrix} = 0 \Rightarrow \lambda^2 - \sqrt{2}\lambda + 1 = 0$$

$$\lambda_1 = \frac{\sqrt{2}+\sqrt{2}j}{2}$$

$$\lambda_2 = \frac{\sqrt{2}-\sqrt{2}j}{2}$$

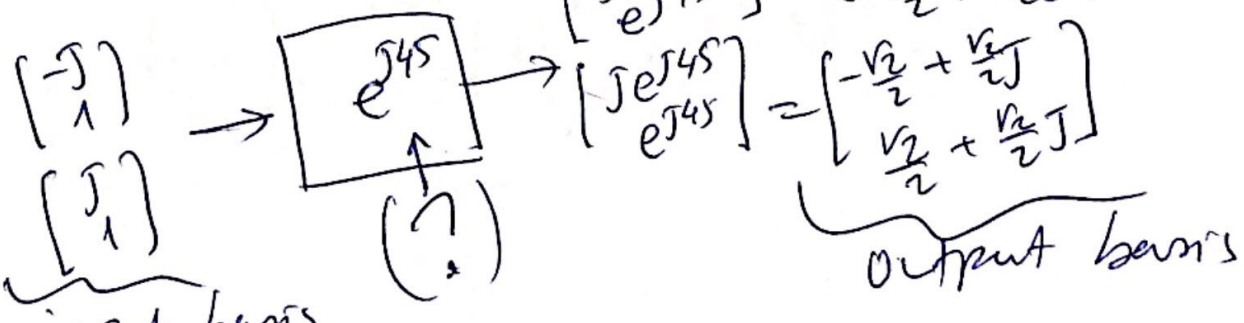
Gerisi 10 puan

$$\lambda_1 = \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}j \Rightarrow \begin{bmatrix} -\frac{\sqrt{2}}{2}j & -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2}j \end{bmatrix} \cdot X_1 = 0 \Rightarrow X_1 = \begin{bmatrix} -j \\ 1 \end{bmatrix}$$

$$\lambda_2 = \frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2}j \Rightarrow \begin{bmatrix} \frac{\sqrt{2}}{2}j & -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2}j \end{bmatrix} \cdot X_2 = 0 \Rightarrow X_2 = \begin{bmatrix} j \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} -je^{j45} \\ e^{j45} \end{bmatrix} = \begin{bmatrix} \frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2}j \\ \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}j \end{bmatrix}$$

$$\begin{bmatrix} je^{j45} \\ e^{j45} \end{bmatrix} = \begin{bmatrix} -\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}j \\ \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}j \end{bmatrix}$$



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$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \cdot \begin{bmatrix} -j & j \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} \frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2}j & -\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}j \\ \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}j & \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}j \end{bmatrix}$$

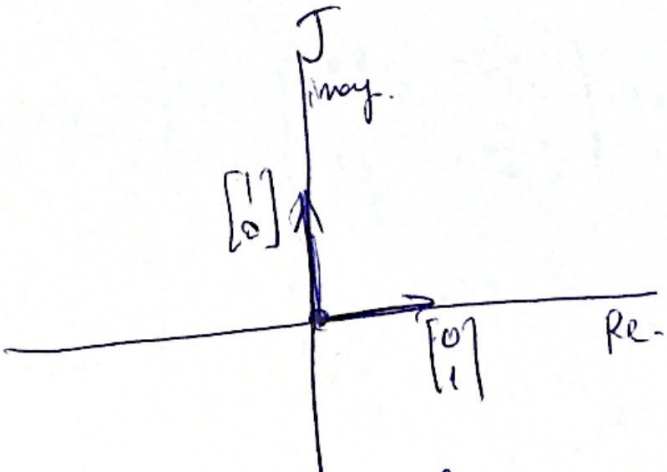
?

$$\left.\begin{aligned} b - aj &= \frac{\sqrt{2}}{2}(1-j) \\ b + aj &= \frac{\sqrt{2}}{2}(-1+j) \end{aligned}\right\} \Rightarrow \begin{aligned} b &= 0 \\ a &= \frac{\sqrt{2}}{2}(-1+j)\cdot\frac{1}{j} \end{aligned}$$

$$a = \frac{\sqrt{2}}{2}(1+j)$$

$$\left.\begin{aligned} d - cj &= \frac{\sqrt{2}}{2}(1+j) \\ d + cj &= \frac{\sqrt{2}}{2}(1+j) \end{aligned}\right\} \Rightarrow \begin{aligned} d &= \frac{\sqrt{2}}{2}(1+j) \\ c &= 0 \end{aligned}$$

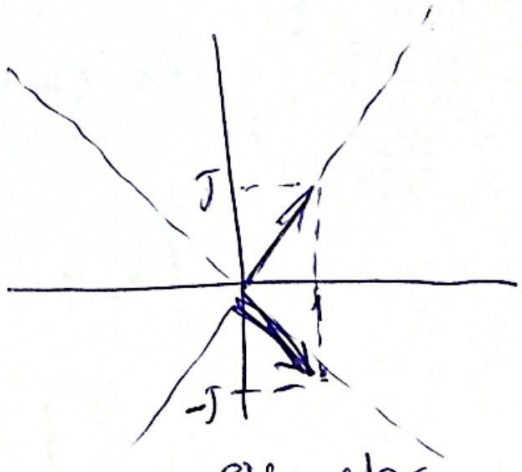
transformation matrix: $\begin{bmatrix} \frac{\sqrt{2}}{2}(1+j) & 0 \\ 0 & \frac{\sqrt{2}}{2}(1+j) \end{bmatrix}$



Natural basis.

$$\begin{bmatrix} \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{bmatrix}$$

transformation matrix for NATURAL BASIS



Eigvecte basis.

$$\begin{bmatrix} \frac{\sqrt{2}}{2}(1+j) & 0 \\ 0 & \frac{\sqrt{2}}{2}(1+j) \end{bmatrix}$$

transformation matrix for EIGENVECTOR basis

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